

Weekly Report 5: Diamond Silhouette and Voxel-Based 3D Reconstruction

Project Number: 18

1 Introduction

This week, we extended our silhouette-based voxel carving pipeline from synthetic shapes to a real diamond model. Silhouette images were considered at **1.8° increments**, resulting in a full set of **200 views** for 360° coverage.

A detailed 3D mesh of a diamond (containing **1944 vertices and 3871 faces**) was used as a reference to validate reconstruction quality. This marks the transition from preliminary testing to carving a real gemstone shape.

2 Methodology

2.1 Voxel Grid Creation

A uniform voxel grid of size **64 × 64 × 64** was initialized in the range $[-1, 1]^3$. This grid size was chosen based on accuracy and available memory resources.

2.2 Diamond Silhouette Sampling

The diamond model was considered from multiple viewpoints spaced at **1.8°**. Each viewpoint produced a binary silhouette defining the visible boundary of the diamond.

2.3 Visual Hull Carving

For each silhouette view:

- Rotate voxel grid by $i \times 1.8^\circ$
- Project voxels to 2D image plane
- Remove voxels lying outside the silhouette mask

Pseudocode:

```
for each view i in range(200):  
    angle = i * 1.8 degrees  
    rotate voxel grid  
    project voxel positions to (u,v)  
    if mask(u,v) == 0: remove voxel
```

2.4 Reference Model Usage

The real diamond mesh was used solely for shape comparison and verification, ensuring that the carved voxel volume matches expected geometric boundaries.

3 Results

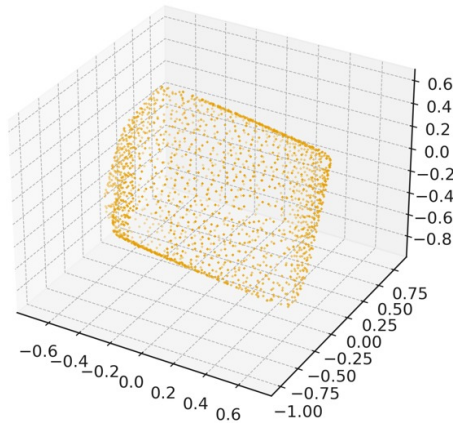


Figure 1: Diamond shape emerging through multi-view voxel carving

Observations:

- Carving from silhouettes produced a recognizable diamond shape
- Facet-like structures began to appear as more views were processed
- Volume closely matched reference geometry

4 Discussion

This experiment confirms successful transition from synthetic shapes to real diamond reconstruction. Voxel resolution and number of silhouettes directly affected detail quality. Future improvements will involve surface smoothing and mesh extraction.

5 Conclusion

This week we:

- Applied voxel carving to a real diamond silhouette dataset
- Used 200 views at 1.8° increments
- Achieved a 3D voxel diamond shape consistent with reference geometry

Next steps include surface reconstruction, refinement, and quantitative evaluation.